

Description of files in this folder

Files associated with intersecting age-associated metabolites with microbe-associated metabolites

1. Darst_Chen_intersection_code.Rmd
 - a. Script inputs:
 - i. Darst_Table_S1.xlsx
 1. Table S1 from <https://doi.org/10.18632/aging.101837>
 2. You can download this file from the link above
 - ii. Modified_Chen_Table_S3_Human_Metabolites_Tab.xlsx
 1. A modified version of the “Human Metabolites” Tab of Table S3 from <https://doi.org/10.1016/j.cell.2023.05.024>
 2. Specifically, we:
 - a. Filtered to just the metabolites with “Bacteria” in the “Source” column, and
 - b. Added (at the end) nine metabolites well-known to be microbially producible and/or producible through host-microbe cometabolism
 3. We have included this file in this folder
 - b. Script outputs:
 - i. Darst_Chen_intersection.txt (available in this folder)
2. menni_fuzzy_matching.py
 - a. Pip-installable packages required: fuzzywuzzy (we used v0.18.0), python-levenshtein, openpyxl, pandas
 - b. Script inputs:
 - i. Menni_Table_S2.xlsx
 1. Excel version of Table S2 from <https://doi.org/10.1093/ije/dyt094>
 2. You can download the non-Excel version from the link above
 - ii. Modified_Chen_Table_S3_Human_Metabolites_Tab.xlsx (available in this folder, see above)

- c. Script outputs:
 - i. Menni_fuzzy_matches_to_microb_metab.txt
 - 1. We have included this file in this folder
 - 2. For each metabolite in the Menni table, this file lists the best fuzzy match to the metabolites in the Chen table
- 3. Menni_fuzzy_matches_to_microb_metab_edited.xlsx
 - a. First tab = highlighted version of Menni_fuzzy_matches_to_microb_metab.txt, after going through it manually
 - i. Highlighted the “actual” matches, as well as “indoleacetate” and “hippurate” since somehow these were not fuzzy-matched to “indoleacetic acid” and “hippuric acid” in the Chen table, but they represent the same metabolites
 - b. Second tab = just the highlighted metabolites
 - i. Replaced the initial matches to indoleacetate and hippurate with indoleacetic acid and hippuric acid, respectively
- 4. Menni_Chén_intersection_code.Rmd
 - a. Script inputs:
 - i. Menni_Table_S2.xlsx (downloadable from online, see above)
 - ii. Modified_Chén_Table_S3_Human_Metabolites_Tab.xlsx (available in this folder, see above)
 - iii. Menni_fuzzy_matches_to_microb_metab_edited.xlsx (available in this folder, see above)
 - b. Script outputs:
 - i. Menni_Chén_intersection.tsv (available in this folder)
- 5. reformatting_dunn.py
 - a. Script inputs:
 - i. Dunn_Supp_Material_2_AGE_tab.xlsx
 - 1. AGE tab of the Excel file “Supplementary Material 2” from <https://doi.org/10.1007/s11306-014-0707-1>, which you can download from the link

2. For our analysis (see Methods), we only used the sections of the table whose titles ended in “_age.g2”
- b. Script outputs:
 - i. reformatted_dunn_supp_mat_2.txt (available in this folder)
6. dunn_fuzzy_matching.py (requires same pip-installable packages as menni_fuzzy_matching.py)
 - a. Script inputs:
 - i. reformatted_dunn_supp_mat_2.txt (available in this folder, see above)
 - ii. Modified_Chen_Table_S3_Human_Metabolites_Tab.xlsx (available in this folder, see above)
 - b. Script outputs:
 - i. Dunn_fuzzy_matches_to_microb_metab.txt (available in this folder)
 1. For each metabolite in the Dunn table, lists the best fuzzy match to the metabolites in the Chen table
7. Dunn_fuzzy_matches_to_microb_metab_edited.xlsx
 - a. First tab = highlighted version of Dunn_fuzzy_matches_to_microb_metab.txt, after going through it manually
 - i. Highlighted the “actual” matches
 - b. Second tab = just the highlighted metabolites
8. Dunn_Chen_intersection_code.Rmd
 - a. Script inputs:
 - i. reformatted_dunn_supp_mat_2.txt (available in this folder, see above)
 - ii. Modified_Chen_Table_S3_Human_Metabolites_Tab.xlsx (available in this folder, see above)
 - iii. Dunn_fuzzy_matches_to_microb_metab_edited.xlsx (available in this folder, see above)
 - b. Script outputs:
 - i. Dunn_Chen_intersection.tsv (available in this folder)

9. Combining_all_intersections.Rmd

a. Script inputs:

- i. Darst_Chén_intersection.txt (available in this folder, see above)
- ii. Menni_Chén_intersection.tsv (available in this folder, see above)
- iii. Dunn_Chén_intersection.tsv (available in this folder, see above)

b. Script outputs:

- i. Darst_Menni_Dunn_Chén_intersection.tsv (available in this folder)

Files associated with narrowing the overlapping metabolites to those most strongly associated with the gut microbiome

1. Darst_Menni_Dunn_Chén_intersection_annotated.xlsx

a. Same as Darst_Menni_Dunn_Chén_intersection.tsv, but annotated with the following information for each metabolite:

- i. r^2 from Supplementary Table 5 from <https://gutsyatlas.serve.scilifelab.se/app/gutsyatlas>
 - 1. NA = not in atlas
 - 2. You can download the table from the link above
- ii. Whether the metabolite is differentially abundant in sera from germ-free vs. conventionally raised mice, from source data for Fig. 5d from <https://doi.org/10.1038/s41467-021-26209-8> (you can download these data from the provided link)

2. choosing_metabolites_for_final_panel.Rmd

a. Script inputs:

- i. Darst_Menni_Dunn_Chén_intersection_annotated.xlsx (available in this folder, see above)
- ii. Supplementary Table 5 from <https://gutsyatlas.serve.scilifelab.se/app/gutsyatlas>
 - 1. You can download the table from the link above

b. Script outputs:

- i. final_panel_metabolites.tsv (available in this folder)

Files associated with making the actual figure (Fig. 1)

3. code_to_make_table_for_fig_1.Rmd

- a. Inputs:
 - i. Darst_Menni_Dunn_Chen_intersection_annotated.xlsx (available in this folder, see above)
 - b. Outputs:
 - i. Darst_Menni_Dunn_Chen_intersection_annotated_dedup_orig.xlsx (available in this folder)
4. Darst_Menni_Dunn_Chen_intersection_annotated_dedup.xlsx
- a. Modified version of
Darst_Menni_Dunn_Chen_intersection_annotated_dedup_orig.xlsx
(available in this folder, see above)
 - b. Modifications compared to original:
 - i. The HMDB_class column was manually added, based on searching each HMDB ID in the online version of HMDB
 - ii. Some of the names in the Original_Met_Name column were adjusted for clarity
 - 1. For example, “glycerol, Glycerol,” which represented two different names for the same metabolite, was changed to simply “glycerol”
 - iii. An “order” column was added to facilitate plotting (next step)
5. code_to_make_fig_1_prelim_version.Rmd
- a. Script inputs:
 - i. Darst_Menni_Dunn_Chen_intersection_annotated_dedup.xlsx (available in this folder, see above)
 - b. Script outputs:
 - i. Figure_1_prelim_version.svg (available in this folder)
 - 1. This file was further modified in Affinity Designer and Adobe Illustrator to yield the final Figure 1